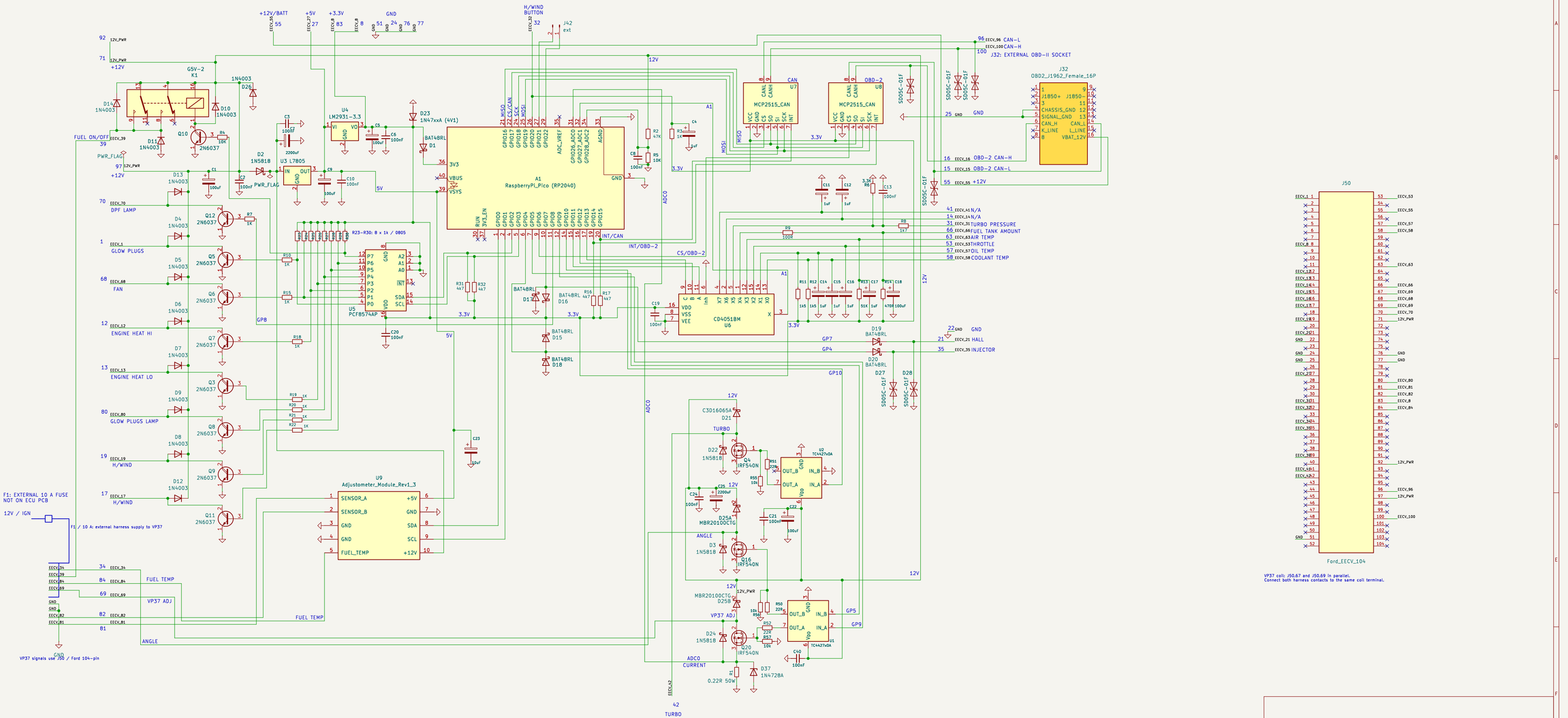

+12V/BATT
+5V
+3.3V
GND
H/WIND BUTTON
J42 ext
12V_PWR
+12V
G5V-2 K1
1N4003
FUEL ON/OFF
PWR_FLAG
U3 L7805
U4 LM2931-3.3
BAT48RL
D1
D23 1N47xxA (4V1)
A1 RaspberryPi_Pico (RP2040)
3V3
VBUS
VSYS
RUN
3V3_EN
ADC_VREF
AGND
INT/CAN
INT/OBD-2
CS/OBD-2
MISO
MOSI
ADC0
U7 MCP2515_CAN
U8 MCP2515_CAN
OBD-2
SD05C-01F
J32 OBD2_J1962_Female_16P
J1850+
J1850-
CHASSIS_GND
SIGNAL_GND
CAN_H
CAN_L
K_LINE
L_LINE
VBAT_12V
CD4051BM U6
U5 PCF8574AP
R23–R30: 8 x 1k / 0805
DPF LAMP
GLOW PLUGS
FAN
ENGINE HEAT HI
ENGINE HEAT LO
GLOW PLUGS LAMP
H/WIND
H/WIND
Q10 2N6037
Q11 2N6037
Q12 2N6037
Q5 2N6037
Q6 2N6037
Q7 2N6037
Q3 2N6037
Q8 2N6037
Q9 2N6037
GP8
U9 Adjustometer_Module_Rev1_3
SENSOR_A
SENSOR_B
FUEL_TEMP
+5V
SDA
SCL
F1: EXTERNAL 10 A FUSE NOT ON ECU PCB
12V / IGN
F1 / 10 A: external harness supply to VP37
FUEL TEMP
VP37 ADJ
ANGLE
VP37 signals use J50 / Ford 104-pin
J32: EXTERNAL OBD-II SOCKET
OBD-2 CAN-H
OBD-2 CAN-L
+12V
N/A
N/A
TURBO PRESSURE
FUEL TANK AMOUNT
AIR TEMP
THROTTLE
OIL TEMP
COOLANT TEMP
HALL
INJECTOR
GP7
GP4
GP10
GP5
GP9
C3D16065A D21
TURBO
D22 1N5818
Q4 IRF540N
D25A MBR20100CTG
ANGLE
D3 1N5818
Q16 IRF540N
MBR20100CTG D25B
VP37 ADJ
D24 1N5818
Q20 IRF540N
ADC0 CURRENT
D37 1N4728A
0.22R 50W
U2 TC4427xOA
U1 TC4427xOA
J50
Ford_EECV_104
VP37 coil: J50.67 and J50.69 in parallel. Connect both harness contacts to the same coil terminal.
PCB2: R / ceramic C = 0805; electrolytics = radial THT / 25V
CAN and R1: measured footprints; R1 mounting follows accepted mechanical baseline
TC4427: non-inverting, LOW = OFF; 12V_PWR supply.
Marcin Kielesiński PROFI-SOFT
Sheet: /
File: ECU.kicad_sch
Title: Fiesta VP37 ECU
Size: User
Date: 2026-11-09
KiCad E.D.A. 9.0.9
Rev: v3.1-PCB3
Id: 1/1